EECE 571Q

Lecture 03: *Just stay still!*

Tuesday 19 May 2026

Announcements

- Quiz 1 today
- Projects: aiming for 4 groups of 2

Last time

We discussed a variety of possible errors and hardware metrics.

We defined quantum error-correcting codes.
Source: Gottesman, Def. 2.1

"A quantum error-correcting code $(U, \mathcal{E})$ (or QECC for short) is a **partial isometry** $U : \mathcal{H}_K \to \mathcal{H}_N$ with the set of **correctable errors** $\mathcal{E}$ (consisting of linear maps $E : \mathcal{H}_N \to \mathcal{H}_M$) with the following property: $\exists$ quantum operation $\mathcal{D} : \mathcal{H}_M \to \mathcal{H}_K$ such that $\forall E \in \mathcal{E},\ \forall |\psi\rangle \in \mathcal{H}_K$,"

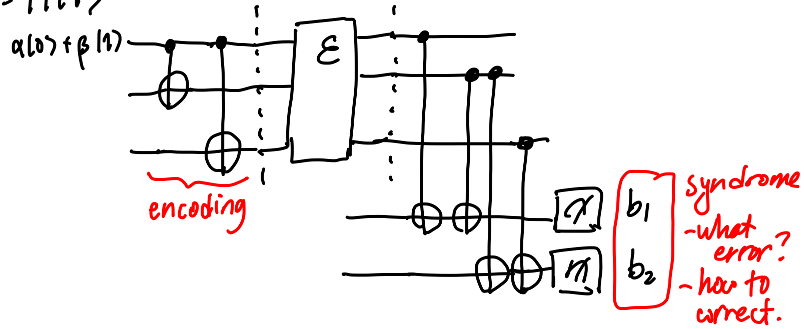
$$\mathcal{D}(EU|\psi\rangle\langle\psi| U^\dagger E^\dagger) = c(E, |\psi\rangle)|\psi\rangle\langle\psi|$$

Last time

We developed a QECC and associated circuitry to correct a single bit flip error, and determined conditions when this was worthwhile.

$|\bar{0}\rangle = |000\rangle$

$|\bar{1}\rangle = |111\rangle$



Learning outcomes

1. Design circuits to detect and correct bit flip, phase flip, and arbitrary errors
2. Outline the conditions under which errors can be corrected
3. Define the Pauli group and state its important properties
4. Express quantum error correcting codes in the stabilizer formalism
5. Characterize whether Pauli errors are correctable by a code using its stabilizer generators

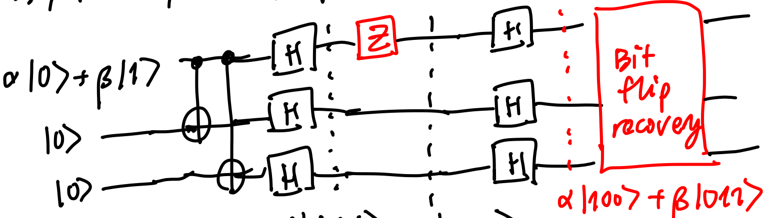
Phase flip code: encoding circuit

Main idea: make phase flip errors “look like” bit flip errors.

$\alpha|000\rangle + \beta|111\rangle \Rightarrow \alpha|000\rangle - \beta|111\rangle$

$|0\rangle / |1\rangle$ Bit flip errors

$|+\rangle / |-\rangle$ phase flip errors



$\alpha|+++\rangle + \beta|---\rangle$

$\alpha|-++\rangle + \beta|+--\rangle$

Conditions for quantum error correction

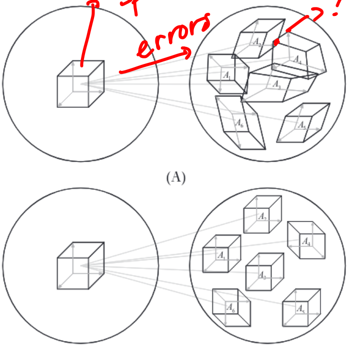
How do we know if a particular error is correctable by a QECC?

Sufficient conditions: non-degenerate code; errors map to orthogonal subspaces.

encoded space

errors

??



(A)

(B)

000 100
001 101
010 110
011 111

X on 1 X on 2

Image: Nielsen and Chuang

Conditions for quantum error correction

$E_b = IIX$

$E_a = XII$

$\langle 111 | E_a^\dagger E_b | 000 \rangle = \langle 011 | 001 \rangle$
$= 0$
$= \langle 111 | 000 \rangle$

Formal statement:

"$(Q, \mathcal{E})$ is a QECC iff $\forall |\psi\rangle, |\phi\rangle \in Q, \forall E_a, E_b \in \mathcal{E}$,

$$\langle \psi | E_a^\dagger E_b | \phi \rangle = \underline{\underline{C_{ab}}} \langle \psi | \phi \rangle$$

Note that C_{ab} does not depend on $|\psi\rangle$ or $|\phi\rangle$" matrix

$\alpha |000\rangle + \beta |111\rangle = |\psi\rangle$

Source: Gottesman, Thm. 2.7

$P = |000\rangle\langle 000| + |111\rangle\langle 111|$ $\quad \gamma |000\rangle + \delta |111\rangle = |\phi\rangle$

P : proj. on codespace $\Rightarrow$ $P E_i E_j^\dagger P = \alpha_{ij} P$ N&C 10.41

Correcting arbitrary errors

Fact: QECCs are linear.

$$\mathcal{D}(EU|\psi\rangle\langle\psi|U^\dagger E^\dagger) = c(E, |\psi\rangle)|\psi\rangle\langle\psi|$$

If E_1 and E_2 and correctable, then

$$\alpha E_1 + \beta E_2$$

is correctable. (Supplementary exercise; see Gottesman 2.4)

Consequence: if we can correct Pauli errors $\mathcal{E} = \{I, X, Z, XZ\}$, we can correct *arbitrary* errors.

Correcting arbitrary errors

Bit flip, phase flip: obviously Paulis

Coherent errors?, e.g., slight over-rotation by angle θ.

$$RX(\varepsilon) = \cos\frac{\varepsilon}{2} I + i\sin\frac{\varepsilon}{2} X$$

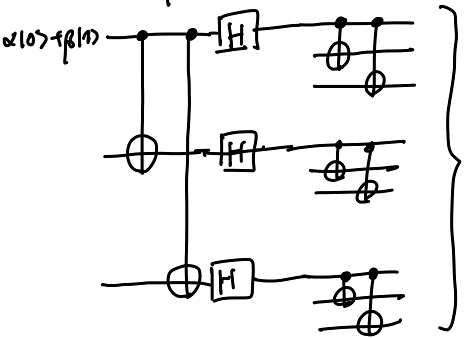
Amplitude damping? *See supplementary exercises*

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix} \quad K_1 = \begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix}$$

Shor code

To correct a combination of one bit flip and/or phase flip error, we can *concatenate* codes.

$$\alpha|0\rangle + \beta|1\rangle \longrightarrow \alpha|{+}{+}{+}\rangle + \beta|{-}{-}{-}\rangle$$



$$|0\rangle_L = \frac{\underbrace{\overset{|\bar{0}\rangle_B+|\bar{1}\rangle_B}{(|000\rangle+|111\rangle)}}_{|\bar{+}\rangle_P}\overset{|\bar{0}\rangle_B+|\bar{1}\rangle_B}{(|000\rangle+|111\rangle)}\overset{|\bar{0}\rangle_B+|\bar{1}\rangle_B}{(|000\rangle+|111\rangle)}}{2\sqrt{2}}$$

$$|1\rangle_L = \underbrace{|\bar{-}\rangle_P}_{|\bar{0}\rangle_B-|\bar{1}\rangle_B}|\bar{-}\rangle_P|\bar{-}\rangle_P$$

Shor code

The Shor code can correct one *arbitrary* error *on a single qubit*.

$$|0\rangle_L = \frac{(|000\rangle+|111\rangle)(|000\rangle+|111\rangle)(|000\rangle+|111\rangle)}{2\sqrt{2}}$$

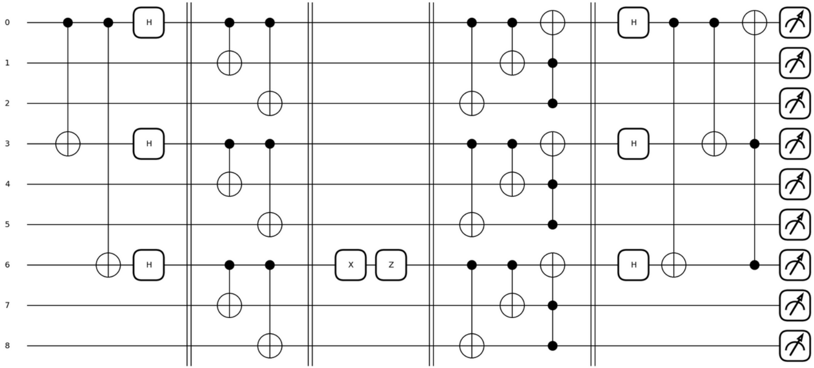
$$|1\rangle_L = \frac{(|000\rangle-|111\rangle)(|000\rangle-|111\rangle)(|000\rangle-|111\rangle)}{2\sqrt{2}}$$

~~Imagine a bit + phase flip on qubit 6:~~

Z on Q5 (or 6 w/ 1 index)

$\Rightarrow$ same outcome as Z on Q4

Shor code



What about errors on *multiple* qubits?

Source: Gottesman 1.3.2-1.3.4 (see 1.3.4 for proofs)

One option: model as *independent channels*, i.e.,

$$\mathcal{E} = \bigotimes_{i=1}^{n} \mathcal{E}_i$$

Revisit our bit flip channel...

$$\Phi_{BF}(\rho) = (1-p)\rho + p X \rho X$$

If we had $\Phi_{BF} \otimes \Phi_{BF} \otimes \Phi_{BF}$ and p is small, many terms contribute very little.

$$p^2 X_0 X_1 \rho X_0 X_1$$

Often more convenient to model using t-qubit error channels.

What about errors on *multiple* qubits?

Source: Gottesman 1.3.3 - see section for proofs

$X \otimes I \otimes Z \otimes I$

A linear operator has **weight** t if it acts non-trivially on t qubits tensored with identity on all other qubits.

A linear operator is a t-**qubit error** if it has the form

$$E = \sum_i E_i$$

and each E_i has weight at most t (not necessarily on same qubits).

An n-qubit channel is a t-**qubit error channel** if it has a Kraus decomposition where all Kraus operators are t-qubit errors.

Code distance

The *distance* of a code is the minimum weight, d, of an error F such that the QECC conditions are violated, i.e.,

$$\langle \psi | \underbrace{F}_{E_a^\dagger E_b} | \phi \rangle \neq c(F) \langle \psi | \phi \rangle$$

(for all possible pairs of codewords).

If E_a, E_b are weight t errors, F has weight at most $2t$.

Consequence: a distance d code

- corrects up to $\lfloor (d-1)/2 \rfloor$ errors
- detects up to $d-1$ errors

Code distance

The Shor code is an $((n, k, d)) = ((9, 1, 3))$ code.

Example: find a minimum-weight F error for 9-qubit code that doesn't satisfy criteria.

$$\underbrace{E_a^\dagger E_b}$$

The Pauli group

The n-qubit Pauli group is

$$\mathcal{P}_n = \{I, X, Y, Z\}^{\otimes n} \times \{1, -1, i, -i\}$$

We often ignore the phases:

$$\hat{\mathcal{P}}_n = \{I, X, Y, Z\}^{\otimes n}$$

We'll use shorthand: $X \otimes Z \otimes I \otimes Y = X_0 Z_1 Y_3$.

Exercise: how many elements are in $\mathcal{P}_n$ and $\hat{\mathcal{P}}_n$?

4^{n+1} $\qquad$ 4^n

Pauli group

The Pauli group is my favourite group. It has many cool properties.

Fact: any $P_1, P_2 \in \mathcal{P}_n$ either commute or anticommute

$[P_1, P_2] = 0$

$= P_1 P_2 - P_2 P_1$

$\{P_1, P_2\} = P_1 P_2 + P_2 P_1$

$= 0$

Fact: the eigenvalues of all Paulis are ± 1. Moreover, the eigenspaces of n-qubit Paulis (except I) each have dimension 2^{n-1}.

Prove in supplementary exercises.

Stabilizers

Let T be a QECC. The **stabilizer** of the code, $S(T)$, is

$$S(T) = \{ M \in \mathcal{P}_n \mid M|\psi\rangle = |\psi\rangle \ \forall |\psi\rangle \in T \}$$

In words: a (sub)group of Paulis for which all codewords are $+1$ eigenstates.

Exercise: Which two-qubit Paulis stabilize this Bell state?

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \implies \begin{array}{c} ZZ \\ XX \end{array}$$

Stabilizers

We will start here Thursday.

Facts about $S(T)$:

- $-I$ is never in $S(T)$
- all elements of $S(T)$ commute (Abelian subgroup of $\mathcal{P}_n$)
- choosing a subgroup S defines the stabilized codewords of T

Stabilizer codes

More convenient to work "backwards": given S, define codespace

If $T = \mathcal{T}(S(T))$, T is a **stabilizer code**.

Bit flip code revisited

An alternative way of thinking about the parity computation is in terms of eigenvalues w.r.t. the code's stabilizer group.

Consider our logical states:

Which three-qubit Paulis are these states $+1$ eigenstates of?

Bit flip code revisited

Let's determine a *minimal* representation of the group in terms of its **stabilizer generators**.

Exercise: if there are r stabilizer generators, how large is the stabilizer group S?

Stabilizers generators

We can also work backwards to obtain codewords from the stabilizer generators by defining a projector onto the codespace,

Π_S provides insight into size of the stabilized space: each generator cuts space in half, so r generators gives space with dimension 2^{n-r}.

Special case: if $r = n$, have a single state called a *stabilizer state*.

Bit flip code: stabilizers and logical operations

Logical Z: need a Pauli that

- stabilizes the logical state but is *not* in S
- *commutes* with everything in S
- has the right action on the subspace

Logical X: need a Pauli that

- stabilizes the logical state but is *not* in S
- *commutes* with everything in S
- has the right action on the subspace
- anticommutes with logical Z

Phase flip code: stabilizer formalism

Stabilizer measurement and error detection

We can use this formalism to construct circuits for error detection and recovery: simply *measure the stabilizer generators*.

Correctible errors anticommute with at least one generator. We can detect their presence in the syndrome measurement.

Bit flip code: stabilizers and logical errors

Stabilizers and logical errors

Danger: error that *commutes* with all elements of S, but isn't in S.

More precisely, define the *normalizer* $N(S)$ as

For an arbitrary error E...

- if $E \notin N(S)$, E is detectable
- if $E \in N(S)$ and $E \in S$, the code space is unchanged
- if $E \in N(S)$ but $E \notin S$, E is **undetectable**

Stabilizers and logical errors

Gottesman Thm 3.4

"The set of undetectable errors for a stabilizer code S is $\hat{N}(S)\backslash\hat{S}$. The **distance** of S is $\min\{\mathrm{wt}(E)|\hat{E} \in \hat{N}(S)\backslash\hat{S}\}$".

Gottesman Thm 3.5

"The stabilizer code S corrects a set of errors $\mathcal{E} \subseteq \mathcal{P}_n$ iff $\hat{E}^\dagger\hat{F} \notin \hat{N}(S)\backslash\hat{S}$ for all $E, F \in \mathcal{E}$".

Shor code: stabilizer formalism

Stabilizer code usually described by notation $[[n, k, d]]$. The 9-qubit Shor code is a [[9, 1, 3]] code.

Name	Operator
g_1	$Z\,Z\,I\,I\,I\,I\,I\,I\,I$
g_2	$I\,Z\,Z\,I\,I\,I\,I\,I\,I$
g_3	$I\,I\,I\,Z\,Z\,I\,I\,I\,I$
g_4	$I\,I\,I\,I\,Z\,Z\,I\,I\,I$
g_5	$I\,I\,I\,I\,I\,I\,Z\,Z\,I$
g_6	$I\,I\,I\,I\,I\,I\,I\,Z\,Z$
g_7	$X\,X\,X\,X\,X\,X\,I\,I\,I$
g_8	$I\,I\,I\,X\,X\,X\,X\,X\,X$
$\bar{Z}$	$X\,X\,X\,X\,X\,X\,X\,X\,X$
$\bar{X}$	$Z\,Z\,Z\,Z\,Z\,Z\,Z\,Z\,Z$

What is an example of a weight-3 undetectable error?

Screenshot: Nielsen & Chuang, chapter 10.5.6.

Syndromes and decoding

Exercise: What are the error syndromes for Y_1 and $X_0 Z_1 X_2$?

Name	Operator
g_1	$Z\,Z\,I\,I\,I\,I\,I\,I\,I$
g_2	$I\,Z\,Z\,I\,I\,I\,I\,I\,I$
g_3	$I\,I\,I\,Z\,Z\,I\,I\,I\,I$
g_4	$I\,I\,I\,I\,Z\,Z\,I\,I\,I$
g_5	$I\,I\,I\,I\,I\,I\,Z\,Z\,I$
g_6	$I\,I\,I\,I\,I\,I\,I\,Z\,Z$
g_7	$X\,X\,X\,X\,X\,X\,I\,I\,I$
g_8	$I\,I\,I\,X\,X\,X\,X\,X\,X$
$\bar{Z}$	$X\,X\,X\,X\,X\,X\,X\,X\,X$
$\bar{X}$	$Z\,Z\,Z\,Z\,Z\,Z\,Z\,Z\,Z$

Screenshot: Nielsen & Chuang, chapter 10.5.6.

Syndromes and decoding

Two errors, E and F are in the same **coset** of $N(S)$ if $F = EN$ for some $N \in N(S)$. Errors in the same coset of $N(S)$ have the same error syndrome.

We will explore this more when we discuss decoding.

Next time

Next class:

- Classical linear codes
- CSS codes
- The Steane code

Action items:

1. Start thinking about project topics
2. Supplementary exercises

Recommended reading:

- From this class: Gottesman Ch. 3; Codebook EC; N&C 10.5
- For next class: Gottesman Ch. 4; N&C 10.4